MIL-C-10113C(EA)
16 September 1974
SUPERSEDING
MIL-C-10113B(MU)
20 July 1967

MILITARY SPECIFICATION

COVER , OUTLET VALVE (FOR C15 OUTLET VALVE)

This specification is approved for use by all
Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification describes one type of molded cover for the C15 outlet valve.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

STANDARDS

FEDERAL

FED-STD-123 - Marking for Domestic Shipping (Civil Agencies).
FED=STD-601 - Rubber: Sampling and Testing.

MILITARY

MIL-STD-105 - Sampling Procedures and Tales for Inspection by Attributes.

DRAWINGS

US ARMY ARMAMENT COMMAND

EDGEWOOD ARSENAL

C5-2-598 - Cover, Outlet Valve.
DL136-19-14 - Tester, Puncture, Natural Rubber, Q38.
DL136-19-22 - Tester, Flexibility, Natural Rubber, Q37.

FSC 4240

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PUBLICATIONS

EDGEWOOD ARSENAL DIRECTIVE

No. 618 - Ozone resistance Test of Natural Rubber Material.

(Copies of specifications, standards, drawings, and publications required by the suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials.

3.1.1 Rubber compound. The robber compoud shall be non-blooming, except for slight wax-type bloom, and shall contain no less than 65 percent. by volume, of new natural rubber. The compound shall be free from lead or mineral sulfates.

3.1.2 Antioxidant and accelerators. The rubber compound shall contain only approved antioxidant and accelerators.

3.1.3 Sun-checking retardant. The rubber compound shall contain a sun-checking retardant.

3.1.4 Copper and manganese inhibitor. The rubber compound shall contain a copper and manganese inhibitor.

3.1.5 Coloring materials. The rubber compoud shall be colored by carbon black, as required.

3.2 Manufacture. The cover shall be manufactured in one piece by the conventional process of molding under heat and pressure, in accordance with Drawing C5-2-598, in molds approved by Edgewood Arsenal.

3.2.1 Repair of flaws. Covers with small flaws may be repaired and accepted, provided there is no change in the formulation or surface continuity.

3.2.2 Test slabs. Test slabs, 6 by 6 inches, by 0.075 ± 0.01 inches, shall be made from stock representative of the formulation used for the preproduction sample and from each sublot during regular production of covers. The slabs shall have a cure equivalent to that of the regular production covers (see 6.2).

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3.3 Physical properties. Physical properties shall conform to the requirements shown in table I when tested as specified therein.

Table I. Physical properties

Property	Unaged	After heat aging	Test paragraph
Average tensile strength, psi, min	3000	2550	4.4.4.2
Average ultimate elongation, %, min	600		4.4.4.2
Percent increase (max)		20	
Percent decrease (max)		25	
Average tensile stress, psi at 200% elongation	250 to 450	250 to 450	4.4.4.2
individual specimens, min.	240	240	
Average tear resistance, lb/in, min			4.4.4.2
Die A or B	300	210	
Die C	225	165	
Average tension set at 75% ultimate elongation, % max	20		4.4.4.2
Ozone resistance, min	80		4.4.4.3
Flexibility			4.4.4.4
Units at 75° ± 5°F., min	7.5		
Units retained at -25° ± 5°F., min	4.8		
*Puncture resistance, psi, min	9.5		4.4.4.5

*To be performed on preproduction sample only.

3.4 Preproduction. Prior to the start of regular production, sample covers and test slabs shall be produced in accordance with this specification for examination and tests (see 4.3). Any change in formulation, operating conditions, or molds shall require a new preproduction sample.

3.5 Workmanship. The cover's inner and outer surfaces shall be smooth and free from imperfections, such as pits, bumps, porous areas, cracks, checks, flash, abrasions, or blisters. The rubber shall be compact and free from holes, bubbles, and foreign matter. Ridges of mold parting lines on outer surfaces shall not be considered as flash when closely trimmed. Rips, tears, or punctures shall be considered as damage.

4, QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

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4.1.1 Supplier's responsibility. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performante of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.1.2 Objective evidence. The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.3)
(b) Quality conformance inspection (see 4.4)

4.3 Preproduction inspection.

4.3.1 Sample. Twenty-five covers, approximately the same number from each mold cavity, and thirty-nine test slabs shall be produced using the same methods, materials, equipment and processes as will be used during regular production.

4.3.2 Inspection procedure.

4.3.2.1 Examination. The sample covers shall be inspected for all requirements of the drawings and of this specification.

4,3.2.2 Tests. Test slabs produced in accordance with 3.4 (see 3.2.2) shall be tested in accordance with 4.4.4.

4.3.3 Acceptance/rejection criteria. The preproduction sample shall meet the examination and tests specified by 4.3.2.1 and 4.3.2.2 to be acceptable. The *supplier* shall obtain written approval from the contracting officer before proceeding with regular production.

4.4 Quality conformance inspection.

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4.4.1 Lotting. A regular production lot shall consist of the covers produced in sequence during one month, as indicated by the dots In the medallion on the cover. A sublet shall consist of the covers produced in not more than one work week from one batch of material (see 6.6). A new lot shall be established following the occurrence of my event affecting the homogeneity of the lot or the use of an alternate approved formulation. Acceptably repaired covers from any accepted parent lot may be incorporated into and assume the lot number of current production.

4.4.2 Sampling

4.4.2.1 For examination. Sampling shall be conducted in accordance with MIL-STD-105.

4.4.2.2 For destructive tests. Thirty-nine test slabs (see 3.2.2) shall be submitted with each sublet for testing.

4.4.3 Inspection procedure.

4.4.3.1 For examination. The sample covers from regular production lots shall be examined in accordance with the classification of defects and with MIL-STD-105.

4.4.3.2 For destructive tests. Sample test slabs shall be tested in accordance with 4.4.4. The acceptance number is 0.

4.4.3.3 Classification of defects.

(a) Cover, outlet valve (Dwg C5 -2-598).

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective.
101	Workmanship (see 3.5)
102	Fit incorrect
103	1/8-hole diameter or alignment incorrect
104	Grease or oil contamination
Minor:	AQL 2.5 percent defective.
201	Thickness incorrect
202	Medallion or marking missing, illegible or incorrect
203	Contamination (other than grease or oil)

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4.4.4 Test.

4.4.4.1 Test specimens. Specimens shall not be buffed Or stretched before testing. Twenty specimens, two from each of 10 test slabs, shall be taken for tensile strength, ultimate elongation, and tensile stress testing. Ten specimens shall be tested 'before aging and 10 shall be tested after aging. Six specimens, two from each of three test slabs, shall be taken, without aging, for tension--set testing. Twenty specimens, two from each of 10 test slabs, shall be taken for tear resistance testing. Ten specimens shall be tested before aging and 10 shall be tested after aging. Twelve specimens, two from each of six test slabs, shall be taken for ozone-resistance testing. Eight specimens, two from each of four test slabs, shall be taken for flexibility testing. Six specimens, one from each of six test slabs, shall be taken for puncture resistance testing. The test specimen selected for heat aging shall be aged in an air oven at a temperature of 158° ± 1.8°F for a continuous period of 168 ± 1 hours or at a temperature of 212° ± 1.8°F for 24 ± 1 continuous hours, as specified in FED-STD-601 (see 6.7).

4.4.4.2 Tensile strength, ultimate elongation, tensile stress, tear resistance, and tension set. Tensile strength, ultimate elongation, tensile stress, tear resistance, and tension set shall be determined as specified in FED-STD-601. The angle-type or crescent type die shall be used for tear resistance testing.

4.4.4.3 Ozone resistance. Ozone resistance shall be determined in accordance with Directive No. 618.

4.4.4.4 Flexibility. Flexibility shall be determined using the Q37 Natural Rubber Flexibility Tester (DL136-19-22). Flexibility shall be read on the grid of the apparatus. Four specimens shall be tested at 75° ± 5°F and four specimens shall be tested at -25° ± 5°F after being conditioned for 72 hours at each temperature.

4.4.4.5 Puncture resistance. Resistance to hydrostatic pressure shall be determined using the Q38 Natural Rubber Puncture Tester (DL136-19-14).

5* PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking, interplant shipment (see 6.8). The covers shall be packaged and packed to provide adequate protection from physical damage from the supply source to the first receiving activity for immediate use of further processing. Shipping containers shell be in compliance with the rules and regulations applicable to the mode of transportation. Marking shall be in conformance with FED-STD-123 and shall include the contractors name, address, and contract number.

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5.2 Repair Parts. When this item is being procured for storage and issue as a repair part, preservation, packaging, packing, and marking shall be as specified on the packaging data sheet which is identified by its National Stock Number.

6. NOTES

6.1 Intended use. This specification describes a protective cover intended to be used on the C15 outlet valve,

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Preproduction:

(1) Time allowed for supplier submission of samples for Government test and evaluation after award of contract.
(2) Name and address of test facility and shipping instructions when testing is performed by the Government.
(3) Time required for the Government to notify the supplier whether of not to proceed with production.

6.3 Equivalent cure. Slabs and covers of the same stock and gage are considered to 'have equivalent cures or the same state of vulcanization when:

(a) Their free--sulfur contents do not vary more than 10 percent.
(b) Their average tensile-stress values at any given elongation (other than ultimate elongation) do not vary more than ± 2.5 percent.
(c) Their curing times can be equated in the following equation:

$$t_2 = t_1 \times 1.6^{\frac{T_1 - T_2}{10}}$$

where: t_1 = minutes of cure at temperature T_1,

t_2 = minutes of cure at temperature T_2,

T_1 = temperature (°F) of cure for time t_1, and

T_2 = temperature (°F) of cure for time t_2.

For convenience, the nomograph in figure 1 may be used instead of the mathematical computation time-temperature relationships in determining equivalent cure.

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FOR

NATURAL RUBBER

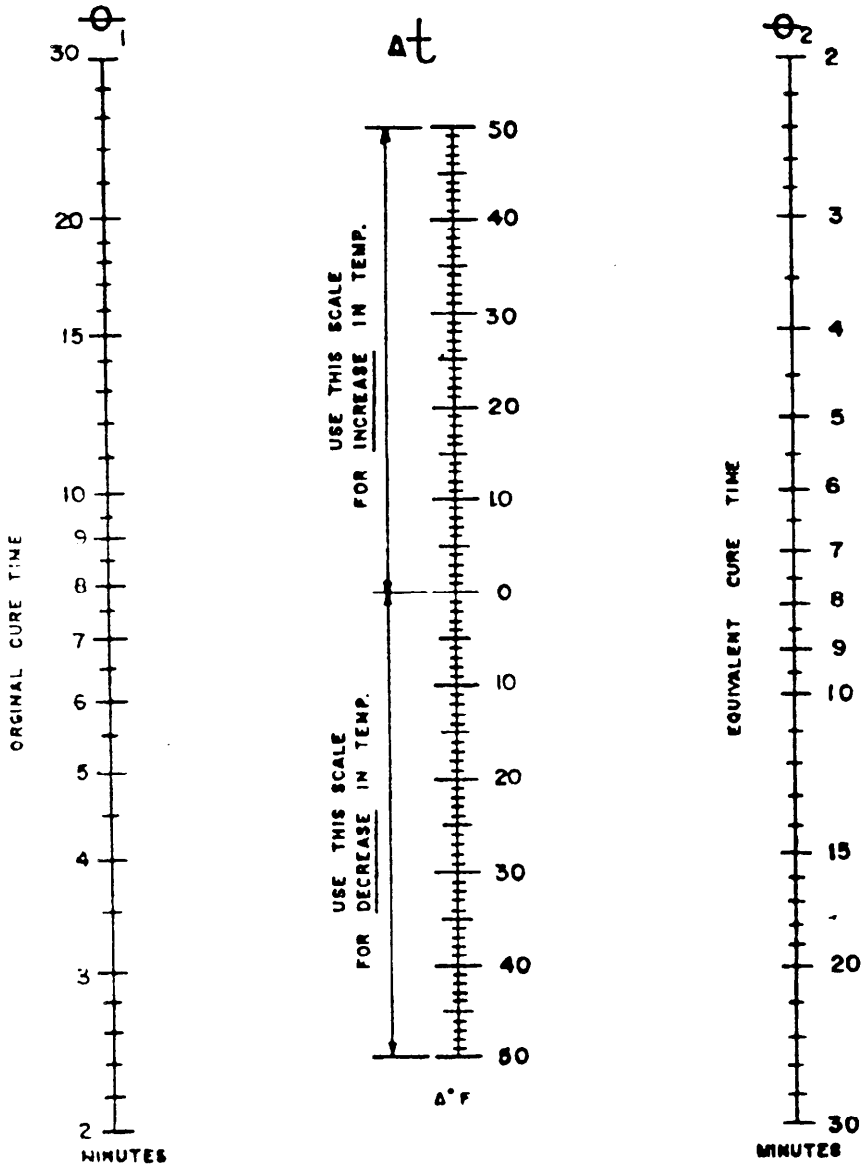


Figure 1.

USING A STRAIGHT EDGE, CONNECT THE ORIGINAL CURE TIME ON θ_1 SCALE WITH THE CHANGE OF CURE TEMPERATURE ON Δt SCALE. EXTEND THE LINE UNTIL IT INTERSECTS θ_2 SCALE AT THE EQUIVALENT CURE TIME.

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6.4 Major change. The term applies to any change in formulae, processing, or curing of currently approved rubber stocks. The major change should be considered effected when one or more of the following conditions occur:

(a) Components of formulae, are varied in excess of the limitations stated in the manufacturers ' tolerance (see 6.5).
(b) The addition or deletion of one or more components of formulae.
(c) Substitutes for one or more components of formulae without advance approval of Edgewood Arsenal.
(d) Deviations from the submitted and approved factory mold cure exceeding +5°F or -10°F, even though equivalent cure is attained.
(e) Use of mold lubricants, dusting powders, or wet sprays not used in preparation of cure of the original samples submitted for approval.

6.5 Manufacturer's tolerances. The manufacturer should submit to Edgewood Arsenal the variations in amounts used, or tolerances, for accelerators, plasticizers, and softeners. These tolerances are permitted to enable the manufacturer to produce stock which will consistly comply with this specification and which will process satisfactorily in the plant. Components of formulae may be submitted on the basis of percentage of stock or number of parts of rubber, and the manufacturer's tolerances recorded in the same units.

6.6 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process and subjected to some physical mixing operation intended to make the final product substantially uniform.

6.7 Aging. Aging at 212° ± 1.8°F for a continuous period of 24 hours is slightly more severe on rubber than aging at 158° ± 1.8°F for 168 consecutive hours; therefore, specimens that fail to meet requirements when aged at the higher temperature may be discarded and replaced by new specimens aged at 158° + 1.8°F for 168 consecutive hours .

6.8 Inter-plant shipments. Packaging for supplies and materials which will not enter the military supply system. Typical inter-plant shipments are shipments from a vendor to a subcontractor or a prime contractor, or from a subcontractor to a prime contractor, or from a vendor or contractor to a military arsenal or plant.

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6.9 International interest. Certain provisions of this specification are the subject of international standardization agreement QSTAG 216 and 234. When amendment, revisions or cancellation of this specification is proposed which will affect or violate the international agreement concerned, the preparing activity will take appropriate reconciliation action through international standardization channels including department standardization offices, if required.

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Preparing activity:

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DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

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